4CS001 –

Introductory Programming

&

Problem-Solving

Lab Report

Name:

Group:

Student ID:

Module Leader

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Tutor

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Table of Contents

Lab No.	Lab Title	Page No.	Tutor's Signature
1			
2			
Lab No.	Lab Title	Page No.	Tutor's Signature

3			
4			

Lab No.	Lab Title	Page No.	Tutor's Signature

5			
6			
Lab No.	**Lab Title**	**Page No.**	**Tutor's Signature**

7			
8	**Exam Week**		

Lab No.	Lab Title	Page No.	Tutor's Signature
9			

10			

Name / Sign of Module Leader **Name / Sign of Module Leader**

_____ _____

Date: **Date:**

Week 1 Questions (*Algorithm and flowchart is mandatory*)

1. Write an Algorithm and produce flowchart to calculate a simple interest rate.
2. Write an Algorithm and flowchart to ask the user to enter a temperature in Celsius and convert it to Fahrenheit.
3. Create flowcharts of Rock, Paper scissors game.
4. Write an Algorithm and flowchart to print numbers in Fibonacci series (numbers less than 10).

Note: Lab report must be handwritten, you will receive weekly questions related to what you have studied throughout the week.

Aims:

Objectives:

Algorithm / Flowchart

Note: For week 1 you don't have to write code.

Write a program to Add two numbers – *refer to solution 1 below* – this is just an example.

***Example* Solution 1**

Aims:

The aim of this lab report is to write a Python program that adds two numbers.

Objectives:

- To understand how to define variables in Python.
- To learn how to use the input function to accept user input in Python.
- To understand how to use the print function to display output in Python.
- To learn how to use the arithmetic operators in Python to perform addition.
- To understand how to write a Python program that adds two numbers and displays the result to the user.

Algorithm / Flowchart

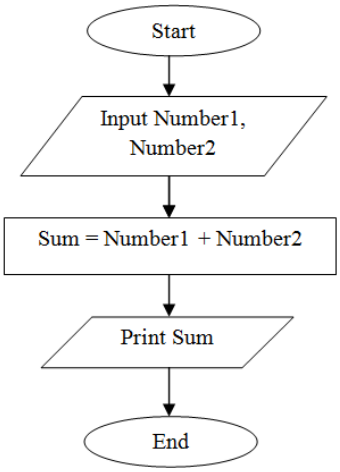
Step 1: Start

Step 2: Enter two numbers in a and b.

Step 3: Add a and b and store in c.

Step 4: Display c

Step 5: End



Source Code:

```
# Prompt the user to enter the first number
num1 = float (input ("Enter the first number: "))
# Prompt the user to enter the second number
num2 = float (input ("Enter the second number: "))
# Add the two numbers together
result = num1 + num2
# Print the result to the user
print ("The sum of", num1, "and" num2, "is", result”)
```

Output

Enter the first number: 5.

Enter the second number: 7.

The sum of 5.0 **and** 7.0 **is** 12.0.

Note: Coding will be started from week 2.